GSC

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1 Solving for the Equilibrium Transition of the Model (Algorithm)

- Let $t = 0, 1, 2, \ldots$. Assume that you know the values of all wedges for $t = 0, 1, \ldots, T^W$, and that the wedges remain unchanged after T^W.
- Fix a large period $T >> T^W > 0$ (e.g. $T = 200$) and assume that the model has reached a steady state at T in which all variables remain constant.
- **Step 1:** Solve for the steady state of the model in period T:
  - **Algorithm:** See Section **??**.
  - **Output:** $K_{i,ss}, E^C_{i,ss}, w_{i,ss}$ and $B_{i,ss} = \bar{B}_i$ for all $i = 1, \ldots, I$.
- **Step 2** Solve for the transition of the model between $t = 0$ and $t = T$, given $K_{i,0}$ and $W_{i,0} \equiv R_0 B_{i,0}$ for $i \in \{US, RW\}$, such that $W_{US,0} + W_{RW,0} = 0$:

  1. Guess sequences $\left\{K^g_{i,t}, E^{Cg}_{i,t}, w^g_{i,t}\right\}_{t=0}^{T}$ for $i \in \{US, RW\}$ such that:

  $$\begin{aligned} K^g_{i,0} &= K_{i,0} \quad \text{and} \quad K^g_{i,T} = K_{i,ss}, \\ E^{Cg}_{i,T} &= E^C_{i,ss}, \\ w^g_{i,T} &= w_{i,ss}. \end{aligned}$$

  and construct implied guess for investment from:

  $$X^g_{i,t} = K^g_{i,t+1} - (1-d)K^g_{i,t}.$$

  2. Use the static equilibrium conditions to obtain the rental rate for capital, $r_{i,t}$, sectoral prices, $P^j_{i,t}$, trade shares, $\pi^j_{ih,t}$, the price index for investment, $P^X_{i,t}$, the price index for consumption, $P^C_{i,t}$, sectoral consumption expenditure shares, $s^j_{i,t}$, Euler equation wedges, $\widetilde{\phi}_{i,t}$, and implied next exports, $\overline{NX}_{i,t}$.
  3. Obtain next exports for all t implied by the budget constraint:

  $$\begin{aligned} NX^g_{US,t} &= w^g_{US,t} L_{US,t} + r_{US,t} K^g_{US,t} - P^X_{US,t} X^g_{US,t} - E^{Cg}_{US,t}, \\ NX^g_{RW,t} &= -NX^g_{US,t}. \end{aligned}$$

  4. Construct implied sequences of $B^g_{US,t+1}$ and R_{t+1} recursively starting at $t = 0$ as follows:
     (a) Obtain $B^g_{i,t+1}$ for t from

  $$\begin{aligned} B^g_{US,t+1} &= NX^g_{US,t} + R_t B^g_{US,t}, \\ B^g_{RW,t+1} &= -B^g_{US,t+1}, \end{aligned}$$

  where $R_0 B^g_{US,0} = W_{US,0}$ for $t = 0$.

(b) Obtain R_{t+1} for t from

$$R_{t+1} = \frac{1}{\delta}\left[\frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}}\frac{E^{Cg}_{US,t}}{1+\phi_{US,t}(B^{g}_{US,t+1}-\bar{B}_{US})} + \frac{\widetilde{\phi}_{RW,t+1}}{\widetilde{\phi}_{RW,t}}\frac{\left(1-E^{Cg}_{US,t}\right)}{1+\phi_{RW,t}-(B^{g}_{US,t+1}-\bar{B}_{US})}\right]^{-1}.$$

(c) Check equilibrium conditions:

$$\frac{R_{t+1}}{1+\phi_{i,t}\left(B^{g}_{i,t+1}-\bar{B}_{i}\right)}P^{X}_{i,t} = \frac{\chi_{i,t+1}}{\chi_{i,t}}\left[r_{i,t+1}+(1-d)P^{X}_{i,t+1}\right], \tag{1}$$

$$\frac{E^{Cg}_{i,t+1}}{E^{Cg}_{i,t}} = \frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}}\frac{\delta R_{t+1}}{1+\phi_{i,t}(B_{i,t+1}-\bar{B}_{i})}. \tag{2}$$

2 Solving for the Steady State of the Model (Algorithm)

- In a steady state of the model, we have that $B_{i,ss} = \bar{B}_i$, and consumption expenditure is constant over time. Therefore, from the Euler equation for bonds, (**??**), we obtain the gross interest rate in steady state:

$$R_{ss} = \frac{1}{\delta}. \tag{3}$$

- Compute the steady state level of net exports:

$$NX_{i,ss} = -(R_{ss} - 1)B_{i,ss}. \tag{4}$$

- Compute the steady state levels of capital, $K_{i,ss}$, and wages, $w_{i,ss}$, by following these steps:
  - **Step 1: (start of outer loop)** Guess a level of capital stocks, $K_{i,ss}$.
  - **Step 2: (start of inner loop)** Guess wages, $w_{i,ss}$.
    1. Compute the rental rate of capital, $r_{i,ss}$, from (**??**).
    2. Compute sectoral prices, $P^j_{i,ss}$, by solving (**??**) and (**??**) simultaneously.
    3. Compute sectoral trade shares, $\pi^j_{ih,ss}$, from (**??**).
    4. Compute the price of investment, $P^X_{i,ss}$, from (**??**).
    5. Compute the steady state level of investment: $X_{i,ss} = dK_{i,ss}$.
    6. Compute the steady state level of consumption expenditure:

$$E^C_{i,ss} = w_{i,ss}L_{i,T} + r_{i,ss}K_{i,ss} - P^X_{i,ss}X_{i,ss} - NX_{i,ss}.$$

    7. Compute the price of consumption, $P^C_{i,ss}$, and level of consumption, $C_{i,ss}$, by solving (**??**) and (**??**) simultaneously.
    8. Compute sectoral levels of consumption expenditure, $E^{C,j}_{i,ss}$, from (**??**) and of investment expenditure, $E^{X,j}_{i,ss}$, from (**??**).
    9. Solve for sectoral levels of expenditure, $E^j_{i,ss}$, and output, $Y^j_{i,ss}$, by solving the system of equations determined by (**??**) and (**??**).
    10. Compute $\overline{NX}_{i,ss}$ from (**??**).
  - **Step 3: (end of inner loop)** Define the error for the inner loop as $\varepsilon^w = NX_{US,ss} - \overline{NX}_{US,ss}$.
    - If $|\varepsilon^w| > tol$ where tol is a small number which determines our tolerance on the error (e.g. 10^{-6}), then compute an update for wages, $w^{up}_{i,ss}$, as follows and go back to **Step 2** with the new guess for wages given by the update:

$$\begin{bmatrix} w^{up}_{US,ss} \\ w^{up}_{RW,ss} \end{bmatrix} = \begin{bmatrix} w_{US,ss}(1-\varepsilon^w) \\ w_{RW,ss}(1+\varepsilon^w) \end{bmatrix}.$$

    - If $|\varepsilon^w| < tol$, then proceed to **Step 4**.
  - **Step 4: (end of outer loop)** Define the error for the outer loop as

$$\varepsilon^K_i = \frac{1}{\delta} - \left(\frac{r_{i,ss}}{P^X_{i,ss}} + (1-d) \right).$$

    - If $\varepsilon^K_i > 0$, then guess of capital is too high.
    - If $\varepsilon^K_i < 0$, then guess of capital is too low.

3 Equilibrium Conditions

- Consider the sequence of budget constraints (already accounting for the governments balanced budget):

$$E^C_{i,t} + P^X_{i,t} X_{i,t} + B_{i,t+1} = w_{i,t} L_{i,t} + r_{i,t} K_{i,t} + R_{i,t} B_{i,t}.$$

3.1 Static Equilibrium Conditions

- **Suppose that in any given period t you know the value of the endogenous variables $K_{i,t}, X_{i,t}$ and $E^C_{i,t}$ (in addition to the value of all the wedges). Then:**

- Wage - rental rate ratio:

$$\boxed{\frac{w_{i,t} L_{i,t}}{r_{i,t} K_{i,t}} = \frac{\varphi_i}{1-\varphi_i}.} \tag{5}$$

- Unit costs:

$$\boxed{c^j_{i,t} = \kappa^j_i \left[\left((r_{i,t})^{\varphi_i} (w_{i,t})^{1-\varphi_i} \right)^{\beta^j_i} \left(\prod_{k \in \mathcal{J}} \left(P^k_{i,t} \right)^{\nu^{j,k}_i} \right)^{1-\beta^j_i} \right],} \tag{6}$$

where

$$\kappa^j_i \equiv \left[\varphi_i^{\varphi_i} (1-\varphi_i)^{1-\varphi_i} \right]^{-\beta^j_i} \left(\beta^j_i \right)^{-\beta^j_i} \left(1-\beta^j_i \right)^{-(1-\beta^j_i)} \prod_{m \in \mathcal{J}} \left(\nu^{j,m}_i \right)^{-(1-\beta^j_i)\nu^{j,m}_i}.$$

- Sectoral prices:
  - Solve system of equations with 6 (2×3) unknowns and 6 (2×3) equations:

$$\boxed{P^j_{i,t} = \Gamma \left[\sum_{h=1}^{I} T^j_{h,t} (c^j_{h,t} \tau^j_{ih,t})^{-\theta} \right]^{-\frac{1}{\theta}},} \tag{7}$$

obtained from

$$P^j_{i,t} = \Gamma \left[\Phi^j_{i,t} \right]^{-\frac{1}{\theta}} \quad \text{and} \quad \Phi^j_{i,t} = \sum_{h=1}^{I} T^j_{h,t} (c^j_{h,t} \tau^j_{ih,t})^{-\theta}.$$

- Trade shares:

$$\boxed{\pi^j_{ih,t} = \frac{T^j_{h,t} (c^j_{h,t} \tau^j_{ih,t})^{-\theta}}{\Phi^j_{i,t}}.} \tag{8}$$

- Price of investment:

$$\boxed{P^X_{i,t} \equiv \left[\sum_{j \in \mathcal{J}} \mu^{x,j}_{i,t} \left(P^j_{i,t} \right)^{1-\zeta} \right]^{\frac{1}{1-\zeta}}.} \tag{9}$$

- Price of consumption and aggregate consumption:

– Solve two systems of equations with 2 unknowns and 2 equations each:

$$\boxed{\begin{aligned} P_{i,t}^{C} &= \left[\sum_{j\in\mathcal{J}} \left(\mu_{i,t}^{c,j}\right) \left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-1)} \left(P_{i,t}^{j}\right)^{1-\psi}\right]^{\frac{1}{1-\psi}}, \qquad (10)\\ C_{i,t} &= \frac{E_{i,t}^{C}}{P_{i,t}^{C}}. \qquad (11)\end{aligned}}$$

- Sectoral consumption expenditure:

$$\boxed{E_{i,t}^{C,j} = s_{i,t}^{j} E_{i,t}^{C},} \qquad (12)$$

where

$$s_{i,t}^{j} = \mu_{i,t}^{c,j} \left(\frac{P_{i,t}^{j}}{P_{i,t}^{C}}\right)^{1-\psi} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^{j}-1)}.$$

- Sectoral investment expenditure:

$$\boxed{E_{i,t}^{X,j} = x_{i,t}^{j} E_{i,t}^{X},} \qquad (13)$$

where

$$\begin{aligned} x_{i,t}^{j} &= \mu_{i,t}^{x,j} \left(\frac{P_{i,t}^{j}}{P_{i,t}^{X}}\right)^{1-\zeta}, \\ E_{i,t}^{X} &= P_{i,t}^{X} \left(K_{i,t+1} - (1-d)K_{i,t}\right). \end{aligned}$$

- Sectoral expenditures and value of output:

– Solve system of linear equations with 12 equations and 12 unknowns:

$$\boxed{\begin{aligned} E_{i,t}^{j} &= E_{i,t}^{C,j} + E_{i,t}^{X,j} + \sum_{k=1}^{J} \left(1-\beta_{i}^{j}\right) \nu_{i}^{kj} Y_{i,t}^{k}, \qquad (14)\\ Y_{i,t}^{j} &= \sum_{h} \pi_{hi,t}^{j} E_{i,t}^{j}. \qquad (15)\end{aligned}}$$

- Aggregate net exports in line with static trade flows:

$$\boxed{\overline{NX}_{i,t} = \sum_{j=1}^{J} NX_{i,t}^{j} = \sum_{j=1}^{J} \left(Y_{i,t}^{j} - E_{i,t}^{j}\right).} \qquad (16)$$

- Aggregate net exports in line with budget constraint:

$$NX_{i,t} = \left(w_{i,t}L_{i,t} + r_{i,t}K_{i,t}\right) - \left(E_{i,t}^{C} + P_{i,t}^{X}\left(K_{i,t+1} - (1-d)K_{i,t}\right)\right) \qquad (17)$$

- **In a static equilibrium (like the steady state), one would have to iterate on the guess for wages until**

$$\overline{NX}_{i,t} = NX_{i,t}. \qquad (18)$$

3.2 Dynamic Equilibrium Conditions

- In a full equilibrium of the model:

$$\frac{E^C_{i,t+1}}{E^C_{i,t}} = \frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}} \frac{\delta R_{t+1}}{1+\phi_{i,t}(B_{i,t+1}-\bar{B}_i)}. \tag{19}$$

- We choose to normalize prices in period t such that:

$$\sum_i^I E^C_{i,t} = 1.$$

- Suppose that you guess $\{E^C_{US,t}\}_{t=0}^{\infty}$. Note that, given our price normalization, the following two conditions must hold in every period t:

$$E^C_{US,t+1} = \frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}} \frac{\delta R_{t+1}}{1+\phi_{US,t}(B_{US,t+1}-\bar{B}_{US})} E^C_{US,t},$$
$$1 - E^C_{US,t+1} = \frac{\widetilde{\phi}_{RW,t+1}}{\widetilde{\phi}_{RW,t}} \frac{\delta R_{t+1}}{1+\phi_{RW,t}-(B_{US,t+1}-\bar{B}_{US})} \left(1 - E^C_{US,t}\right).$$

  Adding these two conditions we see that the equilibrium interest rate must satisfy:

$$R_{t+1} = \frac{1}{\delta}\left[\frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}} \frac{E^C_{US,t}}{1+\phi_{US,t}(B_{US,t+1}-\bar{B}_{US})} + \frac{\widetilde{\phi}_{RW,t+1}}{\widetilde{\phi}_{RW,t}} \frac{\left(1-E^C_{US,t}\right)}{1+\phi_{RW,t}-(B_{US,t+1}-\bar{B}_{US})}\right]^{-1} \tag{20}$$

- Given the guess for wages in every period, we obtain $\{NX_t\}_{t=0}^{\infty}$. Then, given $W_{US,0} \equiv R_0 B_{US,0}$, we can construct $\{B_{i,t+1}\}_{t=0}^{\infty}$ recursively as follows:

$$B_{US,t+1} = NX_{US,t} + R_t B_{i,t}. \tag{21}$$

- We obtain the sequence of equilibrium interest rates $\{R_t\}_{t=0}^{\infty}$.

- The final equilibrium condition to be satisfied is:

$$\frac{R_{t+1}}{1+\phi_{i,t}\left(B_{i,t+1}-\bar{B}_i\right)} P^X_{i,t} = \frac{\chi_{i,t+1}}{\chi_{i,t}}\left[r_{i,t+1} + (1-d)P^X_{i,t+1}\right]. \tag{22}$$